Introduction to Artificial Intelligence

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Assignment: Computer Vision

In this lab session, you will practice Computer Vision with the Kaggle challenge "Digit Recognizer" https://www.kaggle.com/c/digit-recognizer . You will be working in your teams of two students each.

1. Read Sections 7.1 and 7.4 of the book "Applied Artificial Intelligence – An Engineering Approach". Focus on the paragraphs relevant for this course.
2. Answer the quick check questions
3. Download the Kaggle training data, test data and sample submission from https://www.kaggle.com/c/digit-recognizer/data
4. Analyze the dataset.
5. In the Pycharm project for your team, add a new Python package "06_CV_OCR" and add a new Juypter notebook "Kaggle_Digit_Recognizer.ipynb in this package
6. Copy the data files to a subfolder "data"
7. Implement the Kaggle challenge
8. Evaluate your model. You may improve your score by modifying your implementation, e.g., pre-processing, other ML approaches, other ML parameters etc.
9. Submit the submission file generated by your implementation to Kaggle
10. Check your position in the leaderboard.
11. Commit your code to the Git repository
12. Add your team number, Kaggle score (float) and Kaggle position (int) in the shared spreadsheet (link in Moodle), tab "06 CV leaderboard"